

Junrong Yan

Global Analysis on Manifolds and Mathematical Physics

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Employment

2024–Present

Zelevinsky Postdoctoral Fellow

Department of Mathematics, Northeastern University, Boston, MA

Mentor: Maxim Braverman

2022–2024

Boya Postdoctoral Fellow

Beijing International Center for Mathematical Research, Peking University, Beijing, China

Mentor: Xiaobo Liu

Education

2017–2022

Ph.D. in Mathematics

University of California, Santa Barbara, Santa Barbara, CA

Advisor: Xianzhe Dai

2013–2017

B.S. in Mathematics

School of Mathematical Sciences, Nankai University, Tianjin, China

Research Interests

Global analysis; analysis on noncompact manifolds; spectral theory of Dirac operators; mathematics of quantum field theory; comparison geometry of submanifolds.

Publications and Preprints

Published or Accepted

- [1] Xianzhe Dai and **Junrong Yan**, “Witten deformation for noncompact manifolds with bounded geometry,” *Journal of the Institute of Mathematics of Jussieu*, 2023, 38 pages.
- [2] Xianzhe Dai and **Junrong Yan**, “Witten deformation on non-compact manifolds: heat kernel expansion and local index theorem,” *Mathematische Zeitschrift*, 2023, 28 pages.
- [3] Xinxing Tang and **Junrong Yan**, “CY/LG correspondence for Weil–Petersson metrics and tt^* structures,” *Journal of Geometry and Physics*, 2025, 34 pages.
- [4] Fagui Li and **Junrong Yan**, “A first eigenvalue estimate for embedded hypersurfaces in positive Ricci curvature manifolds,” *Differential Geometry and its Applications*, 2026, 10 pages.

- [5] **Junrong Yan**, “Witten deformation for non-Morse functions and gluing formula for analytic torsions,” to appear in *Communications in Analysis and Geometry*, 39 pages.

Preprints

- [1] **Junrong Yan**, “A new analytic proof of gluing formula for analytic torsion forms,” arXiv:2301.02591, 2023, 40 pages.
- [2] Martin Puchol and **Junrong Yan**, “Generalized Morse functions, excision and higher torsions,” arXiv:2407.17100, 2024, 107 pages.
- [3] Xianzhe Dai and **Junrong Yan**, “Weyl law for Schrödinger operators on noncompact manifolds, heat kernel, and Karamata–Hardy–Littlewood theorem,” arXiv:2504.15551, 2025, 28 pages.
- [4] Maxim Braverman, Ahmad Reza Haj Saeedi Sadegh, and **Junrong Yan**, “A gluing formula for the $\mathbb{Z}_2$ -valued index of odd symmetric operators,” arXiv:2505.07094, 2025, 18 pages.
- [5] Minghao Wang and **Junrong Yan**, “Feynman graph integrals on Kähler manifolds,” arXiv:2507.09170, 2025, 49 pages.
- [6] Maxim Braverman, Xianzhe Dai, and **Junrong Yan**, “The classical Weyl law for Schrödinger operators on complete Riemannian manifolds,” arXiv:2605.07200, 2026, 35 pages.
- [7] Minghao Wang and **Junrong Yan**, “Renormalizations in holomorphic field theories on Kähler manifolds,” arXiv:2608.00546, 2026, 82 pages.

Awards and Grants

2023	General Program, China Postdoctoral Science Foundation.
2022	Boya Postdoctoral Fellowship, Peking University.
2018	Wilder Award, University of California, Santa Barbara.

Selected Talks

Aug. 2026	“Graph integrals on Kähler manifolds,” <i>Workshop on Global Analysis on Manifolds</i> , Tianjin.
June 2026	“Graph integrals on Kähler manifolds,” <i>University of Bonn</i> , Bonn.
June 2026	“Graph integrals on Kähler manifolds,” <i>Université Paris-Saclay</i> , Orsay.
Apr. 2026	“Graph integrals on Kähler manifolds,” <i>Texas A&M University</i> , College Station.
Mar. 2026	“Graph integrals on Kähler manifolds,” <i>George Mason University</i> , Washington, DC.
Feb. 2026	“Weyl’s law on complete manifolds,” <i>University of California, Santa Barbara</i> , Santa Barbara.
Feb. 2026	“Graph integrals on Kähler manifolds,” <i>University of California, Irvine</i> , Irvine.
Jan. 2026	“Graph integrals on Kähler manifolds,” <i>Boston University</i> , Boston.

- Dec. 2025 “Graph integrals on Kähler manifolds,” *Wuhan University*, Wuhan.
- June 2025 “Heat kernel expansion, Tauberian theory and Weyl law for Schrödinger operators on noncompact manifolds,” *AMS Special Session on Noncommutative Geometry and Applications*, St. Louis.
- June 2025 “Heat kernel expansion, Tauberian theory and Weyl law for Schrödinger operators on noncompact manifolds,” *Xiamen University*, Xiamen.
- May 2025 “Heat kernel expansion and Weyl’s law for Schrödinger-type operators on noncompact manifolds,” *Tsinghua University*, Beijing.
- Apr. 2025 “Heat kernel expansion and Weyl’s law for Schrödinger-type operators on noncompact manifolds,” *University of Tokyo*, Tokyo.
- Mar. 2025 “Witten deformation for non-Morse functions and gluing formulas,” *Go Fishing 2025—Poisson Geometry Meeting*, St. Louis.
- Dec. 2024 “Towards the comparison of higher torsion invariants,” *Around Local Index Theorem*, Paris.
- Sept. 2024 “Higher torsions and Witten-type deformation,” *Northeastern University*, Boston.
- Sept. 2024 “Witten deformation for non-Morse functions and gluing formulas,” *Harvard University*, Boston.
- Aug. 2024 “Higher torsions and Witten-type deformation,” *Workshop on Global Analysis on Manifolds 2024*, Wuhan.
- Feb. 2024 “Witten deformation for non-Morse functions and higher Cheeger–Müller theorem,” *Conference on Dirac Operators: Analysis and Geometry*, Beijing.
- Nov. 2023 “Witten deformation for non-Morse functions and gluing formulas,” *University of Freiburg*, Freiburg.
- Oct. 2023 “Witten deformation for non-Morse functions and gluing formulas,” *Université Paris-Saclay*, Orsay.
- Mar. 2023 “Witten deformation for non-Morse functions and gluing formulas,” *Chern Institute of Mathematics, Nankai University*, Tianjin.
- Sept. 2022 “Calabi–Yau/Landau–Ginzburg correspondence for tt^* structures,” *Peking University*, Beijing.
- June 2022 “Calabi–Yau/Landau–Ginzburg correspondence for tt^* structures,” *The Chinese University of Hong Kong*, Hong Kong.
- Jan. 2022 “Analytic torsion on noncompact manifolds,” *AMS Special Session on Intersections of Geometric Analysis and Mathematical Physics*, Seattle.
- Jan. 2021 “Heat kernel expansion and local index theorem on noncompact manifolds,” *Peking University*, Beijing.
- Dec. 2020 “Heat kernel expansion and local index theorem on noncompact manifolds,” *Chern Institute of Mathematics, Nankai University*, Tianjin.
- Nov. 2020 “Heat kernel expansion and local index theorem on noncompact manifolds,” *East China Normal University*, Shanghai.
- Apr. 2020 “Witten deformation on noncompact manifolds and mirror symmetry for Landau–Ginzburg models,” *UCSB Differential Geometry Seminar*, Santa Barbara.
- Jan. 2020 “Witten deformation on noncompact manifolds,” *AMS Special Session on Differential Geometry and Global Analysis I: Honoring the Memory of Tadashi Nagano*, Denver.

Nov. 2019 “Witten deformation on noncompact manifolds,” *AMS Special Session on Topics in Global Geometric Analysis*, Los Angeles.

Teaching Experience

Instructor of Record

Northeastern University	<i>MATH 7371: Morse Theory</i> , Fall 2025.
	<i>MATH 2331: Linear Algebra</i> , Fall 2025.
	<i>MATH 1342: Calculus 2 for Science and Engineering</i> , Fall 2024; Spring 2025.
Peking University	<i>Calculus 2</i> , Winter 2024.
UC Santa Barbara	<i>MATH 3B: Calculus with Applications</i> , Summer 2021.

Teaching Assistant

MATH 8: Transitions to Higher Mathematics, Spring 2018; Fall 2018; Fall 2020.

MATH 6B: Vector Calculus, Winter 2019.

MATH 34B: Calculus for Social and Life Sciences, Spring 2019.

MATH 6A: Vector Calculus, Fall 2019; Winter 2020; Summer 2020; Winter 2021; Spring 2021.

Reader

MATH 117: Methods of Analysis, Fall 2017; Fall 2019.

MATH 108B: Introduction to Linear Algebra, Winter 2018.

Undergraduate Mentoring

2021	UCSB Directed Reading Program.
Jiasheng Lin	Morse theory and Floer homology.
Kai Yan and Xinzhe Li	Gauge theory, TQFT, and knot theory.

Academic Service

Paper Referee	<i>SIGMA (Symmetry, Integrability and Geometry: Methods and Applications)</i> ; <i>Journal für die reine und angewandte Mathematik</i> ; <i>Pacific Journal of Mathematics</i> ; <i>Communications in Mathematical Physics</i> ; <i>Journal of Geometric Analysis</i> .
Reviewer	zbMATH Open; MathSciNet.